Sai Tej PARUCHURI

Curriculum Vitae

Mechanical Engineering & Mechanics
Lehigh University, Bethlehem, USA
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Education

2015–2020 **Ph.D. in Mechanical Engineering**, *Virginia Tech*, Blacksburg, VA, USA.
Dissertation title: *Topics on Modeling and Estimation of Linear and Nonlinear Piezoelectric Systems.*
Advisor: *Dr. Andrew Kurdila.*
Research Focus: *Dynamics and Controls; Smart Materials and Adaptive Structures; Data-driven Modeling of Smart Material Systems; Estimation & Control; Nonlinear Systems.*

2019–2020 **M.S. in Mathematics**, *Virginia Tech*, Blacksburg, VA, USA.
Thesis title: *Output Regulation of Systems Governed by Delay Differential Equations: Approximations & Robustness.*
Advisor: *Dr. John Burns.*
Research Focus: *Delay-Differential Equations; Output Regulation; Robust Geometric Control.*

2010–2014 **B.E. in Mechanical Engineering**, *Thiagarajar College of Engineering,*, Madurai, India.

Academic Appointments

Jan 2023 – Ongoing **Research Scientist**, *Plasma Control Laboratory, Mechanical Engineering & Mechanics*, Lehigh University, Bethlehem, PA, USA.
Principal Investigator: *Prof. Eugenio Schuster.*

Nov 2020 – Dec 2022 **Postdoctoral Research Associate**, *Plasma Control Laboratory, Mechanical Engineering & Mechanics*, Lehigh University, Bethlehem, PA, USA.
Principal Investigator: *Prof. Eugenio Schuster.*

Research Interests

- Nonlinear Control; Adaptive Control; Optimal & Robust Control; Nonparametric Control; RKHS Embedding Methods; Adaptive Dynamic Programming; Reinforcement Learning.
- Data-Driven Modeling; Koopman Operator Theory; Online Parametric and Nonparametric Estimation; Physics-Informed Machine Learning; Model Reduction.
- Optimization and Surrogate Modeling of High-Temperature Plasmas; Advanced Control of Magnetically Confined Plasmas; AI-Driven Plasma Solutions.

Publications: Journal Articles

[1] K. Shabbir, B. Leard, Z. Wang, **S. T. Paruchuri**, T. Rafiq, and E. Schuster, "MMMnet: A Neural Network Surrogate for Real-Time Transport Prediction Based on the Updated Multi-Mode Model," *Plasma*, vol. 8, no. 3, 2025.

[2] **S. T. Paruchuri**, I. Ward, N. Rist, V. Graber, H. Al Khawaldeh, Z. Wang, E. Schuster, A. Pajares, and J.-W. Juhn, "Density regulation with disruption avoidance in next-generation tokamaks using a safe reinforcement learning-based controller," *Fusion Engineering and Design*, vol. 216, p. 115064, July 2025.

[3] **S. T. Paruchuri**, A. Pajares, and E. Schuster, "Lyapunov-based nonlinear control of nonautonomous systems with individual input constraints," *Automatica*, vol. 173, p. 111998, Mar. 2025.

[4] S. Morosohk, Z. Wang, **S. T. Paruchuri**, T. Rafiq, and E. Schuster, "Optimal control of

the electron temperature profile in DIII-D using machine learning surrogate models," *Fusion Engineering and Design*, vol. 206, p. 114615, 2024.

[5] B. Leard, Z. Wang, **S. T. Paruchuri**, E. Schuster, and T. Rafiq, "Hybrid model predictive control techniques for safety factor profile and stored energy regulation while incorporating NBI constraints," *Nuclear Fusion*, vol. 64, no. 8, p. 086052, 2024.

[6] S. Niu, A. Bouland, H. Wang, F. Fotiadis, A. Kurdila, A. L'Afflitto, **S. T. Paruchuri**, and K. G. Vamvoudakis, "Convergence Rates of Online Critic Value Function Approximation in Native Spaces," *IEEE Control Systems Letters*, pp. 1–1, 2024.

[7] A. J. Kurdila, **S. T. Paruchuri**, N. Powell, J. Guo, P. Bobade, B. Estes, and H. Wang, "Approximation of discrete and orbital Koopman operators over subsets and manifolds," *Nonlinear Dynamics*, vol. 112, pp. 6291–6327, Apr. 2024.

[8] **S. T. Paruchuri**, A. Pajares, T. Rafiq, and E. Schuster, "Nonlinear control of the minimum safety factor in tokamaks by optimal allocation of spatially moving electron cyclotron current drive," *Fusion Engineering and Design*, vol. 207, p. 114612, Oct. 2024.

[9] X. Song, B. Leard, Z. Wang, **S. T. Paruchuri**, T. Rafiq, and E. Schuster, "Control-Oriented Free-Boundary Equilibrium Solver for Tokamaks," *Plasma*, vol. 7, no. 4, pp. 842–857, 2024.

[10] **S. T. Paruchuri**, V. Graber, A. Pajares, and E. Schuster, "Static actuator-sharing algorithm for concurrent control of multiple plasma properties," *Plasma Physics and Controlled Fusion*, vol. 67, p. 015005, Nov. 2024.

[11] **S. T. Paruchuri**, V. Graber, H. A. Khawaldeh, and E. Schuster, "Dynamic Actuator Allocation via Reinforcement Learning for Concurrent Plasma Control Objectives," *IEEE Transactions on Plasma Science*, vol. 52, pp. 4140–4146, Sept. 2024.

[12] S. Morosohk, Z. Wang, **S. T. Paruchuri**, T. Rafiq, and E. Schuster, "Simultaneous control of the electron temperature and safety factor profiles in DIII-D using model-based optimal control techniques," *Plasma Physics and Controlled Fusion*, vol. 67, p. 015012, Dec. 2024.

[13] B. R. Leard, **S. T. Paruchuri**, T. Rafiq, and E. Schuster, "Fast model-based scenario optimization in NSTX-U enabled by analytic gradient computation," *Fusion Engineering and Design*, vol. 192, p. 113606, 2023.

[14] **S. T. Paruchuri** and E. Schuster, "Nonlinear control of safety factor gradient in tokamaks using spatially variable electron cyclotron current drives," *Fusion Engineering and Design*, vol. 194, p. 113914, 2023.

[15] H. Al Khawaldeh, B. Leard, **S. T. Paruchuri**, T. Rafiq, and E. Schuster, "Model-based linear–quadratic–integral controller for simultaneous regulation of the current profile and normalized beta in NSTX-U," *Fusion Engineering and Design*, vol. 192, p. 113795, 2023.

[16] **S. T. Paruchuri**, Z. Wang, T. Rafiq, and E. Schuster, "Model predictive current profile control in tokamaks by exploiting spatially moving electron cyclotron current drives," *Fusion Engineering and Design*, vol. 192, p. 113796, 2023.

[17] J. Burns, B. Estes, J. Guo, A. Kurdila, **S. T. Paruchuri**, and N. Powell, "Kernel methods for regression in continuous time over subsets and manifolds," *Nonlinear Dynamics*, vol. 111, no. 14, pp. 13165–13186, 2023.

[18] **S. T. Paruchuri**, J. Guo, and A. Kurdila, "Sufficient Conditions for Parameter Convergence Over Embedded Manifolds Using Kernel Techniques," *IEEE Transactions on Automatic Control*, vol. 68, no. 2, pp. 753–765, 2023.

[19] A. Bouland, S. Niu, **S. T. Paruchuri**, A. Kurdila, J. Burns, and E. Schuster, "Rates of Convergence in a Class of Native Spaces for Reinforcement Learning and Control," *IEEE Control Systems Letters*, pp. 1–1, 2023.

[20] J. Guo, M. E. Kepler, **S. T. Paruchuri**, H. Wang, A. J. Kurdila, and D. J. Stilwell, "Adaptive estimation of external fields in reproducing kernel Hilbert spaces," *International Journal of Adaptive Control and Signal Processing*, vol. 36, no. 8, pp. 1931–1957, 2022.

[21] **S. T. Paruchuri**, J. Guo, and A. Kurdila, "Kernel center adaptation in the reproducing kernel Hilbert space embedding method," *International Journal of Adaptive Control and Signal Processing*, vol. 36, no. 7, pp. 1562–1583, 2022.

[22] **S. T. Paruchuri**, J. Guo, and A. Kurdila, "Reproducing kernel Hilbert space embedding for adaptive estimation of nonlinearities in piezoelectric systems," *Nonlinear Dynamics*, vol. 101, no. 2, pp. 1397–1415, 2020.

[23] **S. T. Paruchuri**, V. V. S. Malladi, P. A. Tarazaga, and A. J. Kurdila, "Expanding the teaching of single frequency vibration absorption to broadband attenuation using subordinate oscillator arrays via fettuccine pasta," *Engineering Structures*, vol. 224, p. 111201, 2020.

[24] J. Sterling, **S. T. Paruchuri**, T. J. Ryan, J. Vignola, and A. J. Kurdila, "Subordinate Oscillator Arrays: Physical Design and Effects of Error," *engrxiv preprint https://doi.org/10.31224/osf.io/kpv3r*, 2020.

[25] A. J. Kurdila, J. Guo, **S. T. Paruchuri**, and P. Bobade, "Persistence of excitation in reproducing kernel hilbert spaces, positive limit sets, and smooth manifolds," *arXiv preprint arXiv:1909.12274*, 2019.

[26] J. Guo, S. Dadashi, M. Bender, **S. T. Paruchuri**, N. Powell, Y. Sharma, H. Kurdila, J. W. McGlothlin, and A. J. Kurdila, "Probabilistic error bounds on constraint violation for empirical-analytical Lagrangian models of motion," *Nonlinear Dynamics*, vol. 98, no. 1, pp. 195–213, 2019.

[27] **S. T. Paruchuri**, J. Sterling, V. V. N. S. Malladi, A. Kurdila, J. Vignola, and P. Tarazaga, "Passive piezoelectric subordinate oscillator arrays," *Smart Materials and Structures*, vol. 28, no. 8, p. 085046, 2019.

[28] J. A. Sterling, J. F. Vignola, T. J. Ryan, and **S. T. Paruchuri**, "Analysis of increased damping in arrays of attached resonators," *The Journal of the Acoustical Society of America*, vol. 145, no. 3, pp. 1824–1824, 2019.

Publications: Peer-Reviewed Conference Proceedings

[1] Z. Wang, **S. T. Paruchuri**, and E. Schuster, "Safety Factor Profile Control in EAST via Reinforcement-Learning-Based Model Predictive Control," in *2025 IEEE Conference on Control Technology and Applications (CCTA)*, pp. 832–837, 25-27 Aug. 2025.

[2] N. Powell, **S. T. Paruchuri**, A. Bouland, S. Niu, and A. Kurdila, "Invariance and Approximation of Koopman Operators in Native Spaces," in *2024 American Control Conference (ACC)*, pp. 2871–2878, 2024-07-10/2024-07-12.

[3] Z. Wang, **S. T. Paruchuri**, L. Yang, and E. Schuster, "Safety Factor Profile Regulation via Self-Triggered Model Predictive Control in the EAST Tokamak," in *2024 American Control Conference (ACC)*, pp. 2545–2550, 2024-07-10/2024-07-12.

[4] **S. T. Paruchuri**, A. Pajares, and E. Schuster, "Nonlinear Local Control of the Safety-Factor-Profile Gradient at Moving Spatial Locations in Tokamak Plasmas," in *2023 American Control Conference (ACC)*, pp. 1590–1595, 2023.

[5] J. A. Burns, J. Guo, A. J. Kurdila, **S. T. Paruchuri**, and H. Wang, "Error Bounds for Native Space Embedding Observers with Operator-Valued Kernels," in *2023 American Control Conference (ACC)*, pp. 4796–4801, 2023.

[6] **S. T. Paruchuri**, A. Pajares, and E. Schuster, "Lyapunov-based Current-Profile Feedback Control in Tokamaks with Nonsymmetric Individual Actuator Saturation," in *2023 American Control Conference (ACC)*, pp. 1377–1382, 2023.

[7] **S. T. Paruchuri** and E. Schuster, "Distributed Regulation of the Safety Factor Profile in Tokamaks Using Nonlinear Infinite-dimensional Control," in *22nd IFAC World Congress*, vol. 56, pp. 4466–4471, 2023.

[8] **S. T. Paruchuri**, A. Pajares, and E. Schuster, "Minimum Safety Factor Control in Tokamaks via Optimal Allocation of Spatially Moving Electron Cyclotron Current Drive," in *2021 60th IEEE Conference on Decision and Control (CDC)*, pp. 454–459, 2021.

[9] E. Aulisa, J. A. Burns, D. S. Gilliam, and **S. T. Paruchuri**, "Accurate Approximate Regulation of Nonlinear Delay Differential Control Systems," in *2021 60th IEEE Conference on Decision and Control (CDC)*, pp. 422–427, 2021.

[10] J. A. Burns, **S. T. Paruchuri**, and M. Schmidt, "Output Regulation of Systems Governed by Delay Differential Equations: Approximations and Robustness," in *24th International Symposium on Mathematical Theory of Networks and Systems MTNS 2020*, vol. 54, pp. 422–427, 2021.

[11] J. Guo, **S. T. Paruchuri**, and A. J. Kurdila, "Persistence of Excitation in Uniformly Embedded Reproducing Kernel Hilbert (RKH) Spaces," in *2020 American Control Conference (ACC)*, pp. 4539–4544, 2020.

[12] J. Guo, **S. T. Paruchuri**, and A. J. Kurdila, "Approximations of the Reproducing Kernel Hilbert Space (RKHS) Embedding Method over Manifolds," in *2020 59th IEEE Conference on Decision and Control (CDC)*, pp. 1596–1601, 2020.

[13] **S. T. Paruchuri**, J. Guo, M. Kepler, T. Ryan, H. Wang, A. J. Kurdila, and D. Stilwell, "Intrinsic and Extrinsic Approximation of Koopman Operators over Manifolds," in *2020 59th IEEE Conference on Decision and Control (CDC)*, pp. 1608–1613, 2020.

[14] J. Sterling, J. Vignola, J. Gietl, T. Ryan, N. Sonne, and **S. T. Paruchuri**, "Effect of Increased Damping in Subordinate Oscillator Arrays," in *Journal of Physics: Conference Series*, vol. 1149, p. 012006, IOP Publishing, 2018.

[15] **S. T. Paruchuri**, A. Kurdila, and J. Vignola, "Estimation of Distribution Errors in Piezoelectric Subordinate Oscillator Arrays," in *SMASIS2018*, (Volume 1: Development and Characterization of Multifunctional Materials; Modeling, Simulation, and Control of Adaptive Systems; Integrated System Design and Implementation), 2018.

[16] **S. T. Paruchuri**, J. Sterling, A. Kurdila, and J. Vignola, "Piezoelectric composite subordinate oscillator arrays and frequency response shaping for passive vibration attenuation," in *2017 IEEE Conference on Control Technology and Applications (CCTA)*, pp. 702–707, 2017.

[17] **S. T. Paruchuri**, A. J. Kurdila, J. Sterling, A. Vignola, J. Judge, J. Vignola, and T. Ryan, "Thermodynamic Variational Formulations of Subordinate Oscillator Arrays (SOA) With Linear Piezoelectrics," in *IDETC-CIE2017*, (Volume 8: 29th Conference on Mechanical Vibration and Noise), 2017.

Publications: Refereed Conference Abstracts

[1] A. Pajares, H. R. Al Khawaldeh, F. Turco, D. Shiraki, T. Pederson, D. Eldon, J.-W. Juhn, V. R. Graber, E. Schuster, and **S. T. Paruchuri**, "Iter-relevant Density Control with Pellets in DIII-D's ITER Baseline Scenario," in *American Physical Society Division of Plasma Physics Meeting Archives*, (Long Beach, California, USA), APS, 2025.

[2] **S. T. Paruchuri**, H. Al Khawaldeh, V. R. Graber, E. Schuster, A. Pajares, and J.-W. Juhn, "Nonlinear Adaptive Control of Deuterium-Tritium Density Ratio in ITER," in *American Physical Society Division of Plasma Physics Meeting Archives*, (Long Beach, California, USA), APS, 2025.

[3] **S. T. Paruchuri**, H. Al Khawaldeh, V. Graber, Z. Wang, N. Rist, I. Ward, T. Rafiq, E. Schuster, A. Pajares, and J.-W. Juhn, "Assessing Model-based and Learning-based

Strategies for Active Density Regulation in ITER," in *Bulletin of the American Physical Society*, (Atlanta, Georgia, USA), APS, 2024.

[4] K. Shabbir, B. Leard, **S. T. Paruchuri**, T. Rafiq, and E. Schuster, "A Neural Network Version of the Multi-Mode Model for Fast Simulations in DIII-D," in *Bulletin of the American Physical Society*, (Atlanta, Georgia, USA), APS, 2024.

[5] N. Rist, **S. T. Paruchuri**, and E. Schuster, "Reinforcement Learning-Based Control of the Safety Factor Profile and Normalized Beta in Tokamaks," in *Bulletin of the American Physical Society*, (Atlanta, Georgia, USA), APS, 2024.

[6] H. Al Khawaldeh, **S. T. Paruchuri**, V. Graber, T. Rafiq, E. Schuster, A. Pajares, and J.-W. Juhn, "Advanced Density Regulation for ITER-emulated Scenarios on DIII-D via Adaptive Control Techniques," in *Bulletin of the American Physical Society*, (Atlanta, Georgia, USA), APS, 2024.

[7] I. Ward, V. Graber, N. Rist, **S. T. Paruchuri**, and E. Schuster, "Burn Control in ITER Using a Reinforcement Learning Approach," in *Bulletin of the American Physical Society*, (Atlanta, Georgia, USA), APS, 2024.

[8] H. Al Khawaldeh, **S. T. Paruchuri**, T. Rafiq, and E. Schuster, "Model-Based Control of the Dimensionless Gain in KSTAR by Leveraging Real-time Estimation of the Confinement Time," in *Bulletin of the American Physical Society*, (Denver, Colorado, USA), APS, 2023.

[9] Z. Wang, **S. T. Paruchuri**, S. Morosohk, and E. Schuster, "Current Profile Control in EAST via Reinforcement-Learning-based Model Predictive Control," in *Bulletin of the American Physical Society*, (Denver, Colorado, USA), APS, 2023.

[10] S. Morosohk, Z. Wang, **S. T. Paruchuri**, T. Rafiq, and E. Schuster, "Simultaneous Regulation of the Electron Temperature and Safety Factor Profiles for DIII-D using Optimal Control Methods," in *Bulletin of the American Physical Society*, (Denver, Colorado, USA), APS, 2023.

[11] **S. T. Paruchuri** and E. Schuster, "Incorporating Gas Puffing Delays into Density Control Synthesis in Tokamak Reactors By Combining Optimal Control and Reinforcement Learning Techniques," in *Bulletin of the American Physical Society*, (Denver, Colorado, USA), APS, 2023.

[12] B. Leard, Z. Wang, **S. T. Paruchuri**, E. Schuster, and T. Rafiq, "Incorporating Pulse Width Modulation Constraints Arising in Neutral Beam Injection," in *29th IAEA Fusion Energy Conference*, (London, United Kingdom), 2023.

[13] L. Yang, H. Al Khawaldeh, **S. T. Paruchuri**, X. Song, Z. Wang, and E. Schuster, "Towards Density Profile Regulation via Pellet Injection in Tokamaks Using Hybrid Model Predictive Control," in *29th IAEA Fusion Energy Conference*, (London, United Kingdom), 2023.

[14] H. Al Khawaldeh, **S. T. Paruchuri**, Z. Wang, T. Rafiq, and E. Schuster, "Simultaneous Optimal Regulation Of Kinetic+Magnetic Scalar Plasma Properties For Robust Sustainment Of Advanced Scenarios In NSTX-U," in *29th IAEA Fusion Energy Conference*, (London, United Kingdom), 2023.

[15] **S. T. Paruchuri**, V. Graber, A. Pajares, and E. Schuster, "Actuator-Sharing Algorithm For Simultaneous Regulation Of Multiple Plasma Properties With Coupled Dynamics," in *29th IAEA Fusion Energy Conference*, (London, United Kingdom), 2023.

[16] **S. T. Paruchuri** and E. Schuster, "Model-based Safety-Factor-Profile Slope Control at Predefined Rational Surfaces in DIII-D," in *Bulletin of the American Physical Society*, (Spokane, Washington, USA), APS, 2022.

[17] H. Al Khawaldeh, B. Leard, **S. T. Paruchuri**, T. Rafiq, and E. Schuster, "Model-Based Optimal Control of Core Kinetic+ Magnetic Profiles and Scalar Plasma Properties in

NSTX-U," in *Bulletin of the American Physical Society*, (Spokane, Washington, USA), APS, 2022.

[18] H. Wang, J. Burns, J. Guo, A. Kurdila, **S. T. Paruchuri**, and N. Powell, "Koopman Operators and Inverse Problems," in *IEICE Proceedings Series*, vol. 71, The Institute of Electronics, Information and Communication Engineers, 2022.

[19] S. Morosohk, **S. T. Paruchuri**, Z. Wang, T. Rafiq, and E. Schuster, "Robust Control of the Electron Temperature Profile in DIII-D," in *Bulletin of the American Physical Society*, (Spokane, Washington, USA), APS, 2022.

[20] L. Yang, **S. T. Paruchuri**, and E. Schuster, "Electron Density Profile Regulation with Pellet Injection using Self-trigger Model Predictive Control," in *Bulletin of the American Physical Society*, (Spokane, Washington, USA), APS, 2022.

[21] M. Elango, A. Annamalai, and **S. T. Paruchuri**, "Harmonic Analysis on Axial Turbine Blades," in *Recent Advances in Computational and Experimental Mechanics, Vol—I* (D. Maity, P. K. Patra, M. Afzal, R. Ghoshal, C. S. Mistry, P. Jana, and D. K. Maiti, eds.), (Singapore), pp. 23–35, Springer Singapore, 2022.

[22] B. Leard, **S. T. Paruchuri**, T. Rafiq, and E. Schuster, "Towards In-Between-Discharges Model-Based Scenario Planning in NSTX-U Via Fast Nonlinear Optimization," in *Bulletin of the American Physical Society*, vol. 66, (Pittsburgh, Pennsylvania, USA), APS, 2021.

[23] S. Morosohk, Z. Wang, A. Pajares, **S. T. Paruchuri**, T. Rafiq, and E. Schuster, "Simultaneous Electron Temperature and Safety Factor Profile Control for DIII-D," in *Bulletin of the American Physical Society*, vol. 66, (Pittsburgh, Pennsylvania, USA), APS, 2021.

[24] **S. T. Paruchuri**, A. Pajares, T. Rafiq, and E. Schuster, "Control of the Local Gradient and the Minimum Value of the Safety Factor Profile by Using Moving ECCD," in *Bulletin of the American Physical Society*, vol. 66, (Pittsburgh, Pennsylvania, USA), APS, 2021.

[25] C. R. Neighborgall, K. Kothari, V. V. N. Sriram Malladi, P. Tarazaga, **S. T. Paruchuri**, and A. Kurdila, "Shaping the Frequency Response Function (FRF) of a Multi-Degree-of-Freedom (MDOF) Structure Using Arrays of Tuned Vibration Absorbers (TVA)," in *Topics in Modal Analysis & Testing, Volume 8* (M. L. Mains and B. J. Dilworth, eds.), (Cham), pp. 317–326, Springer International Publishing, 2020.

[26] J. Sterling, **S. T. Paruchuri**, P. Tarazaga, J. Vignola, A. Kurdila, V. V. N. S. Malladi, and T. Ryan, "Piezoelectric Subordinate Oscillator Arrays: Performance Recovery and Robustness to Uncertainty," in *IDETC-CIE2019*, (Volume 8: 31st Conference on Mechanical Vibration and Noise), 2019.

[27] J. Vignola, J. Judge, J. Sterling, T. Ryan, A. Kurdila, **S. T. Paruchuri**, and A. Glean, "On the use of shunted piezo actuators for mitigation of distribution errors in resonator arrays," in *Proceedings of the 22nd International Congress on Acoustics, Buenos Aires*, 2016.

Research Proposals and Grants

As Principal Investigator

2023 **Physics-Constrained Data-Driven Estimation and Control Solutions in FPP with Stability and Performance Guarantees**, with Lifang He, Tariq Rafiq, Eugenio Schuster. U.S. Department of Energy (*not funded*).

As Technical Contributor to Proposals

2025 **High-performance Scenario Development in NSTX-U via Physics-based Modeling and AI+Model-based Control**, U.S. Department of Energy (*under review*). PI: Eugenio Schuster, Co-PI: Tariq Rafiq.

2025 **AI-Enhanced, Model-Based, Integrated Control for Safe and Disruption-Free Operation of High-Performance Burning Plasmas in Long-Pulse Tungsten Tokamaks**, U.S. Department of Energy (*under review*). PI: Eugenio Schuster (Lehigh University). Part of a multi-institutional submission.

Research Activities

Experimental Campaigns at Major International Fusion Research Facilities

March 2026 **Proximity Control and Safe Reinforcement Learning Algorithms for Plasma Regulation with Stability Limit Avoidance**, *KSTAR (Korea Superconducting Tokamak Advanced Research)*, Korea Institute of Fusion Energy (KFE), Daejeon, South Korea.

- Led the development and integration of the safe reinforcement learning category in the KSTAR plasma control system (PCS).
- Collaborated with GA and KFE specialists on proximity control and plasma density regulation.
- Led the experimental evaluation of reference governor and safe reinforcement learning algorithms for plasma regulation with density-limit avoidance in KSTAR.

March 2026 **Adaptive Density Regulation for ITER and Future Reactors by Simultaneous Gas Puffing and Pellet Injection**, *KSTAR (Korea Superconducting Tokamak Advanced Research)*, Korea Institute of Fusion Energy (KFE), Daejeon, South Korea.

- Participated in collaborative KSTAR experiments on real-time plasma density control using a controller framework adapted from prior DIII-D studies.
- Provided technical guidance on modifications to the underlying control code during its porting to the KSTAR plasma control system.

Dec 2025 **Optimal Actuator Allocation for Integrated Control**, *TCV, Swiss Plasma Center*, École Polytechnique Fédérale de Lausanne, Switzerland.

- Provided technical and algorithmic guidance for the implementation of an actuator allocation algorithm in the TCV plasma control system in collaboration with the Swiss Plasma Center team and ITER researchers.
- Participated in experimental validation of the allocation algorithm across multiple plasma control test cases.

April 2025 **Advanced Density Control in DIII-D for ITER and Beyond**, *DIII-D National Fusion Facility*, General Atomics, San Diego, CA, USA.

- Guided the development of a model reference adaptive plasma density controller incorporating gas-puffing delays.
- Provided theoretical oversight for the controller design.
- Participated in collaborative experiments demonstrating adaptive plasma-density regulation using pellet injection and delayed gas puffing.

Jan 2025 **Advanced Density Regulation for ITER and Future Reactors by Simultaneous Gas Puffing and Pellet Injection**, *KSTAR (Korea Superconducting Tokamak Advanced Research)*, Korea Institute of Fusion Energy (KFE), Daejeon, South Korea.

- Assisted in the development and validation of KSTAR plasma control system (PCS) infrastructure for mimicking ITER-like delayed gas puffing.
- Collaborated with KFE specialists on plasma control and pellet injection systems.

Teaching, Supervision and Mentoring Experience

Instructor & Lecturer Roles

Spring 2025 **Course Instructor**, *ME 207 Mechanical Engineering Laboratory (Controls Section)*, Lehigh University.

- Number of students: 41 (Batch 1 - 20 students; Batch 2 - 21 students).
- Taught two back-to-back lab batches combining control theory lectures and hands-on experiments.
- Adapted instruction for students with diverse backgrounds; 65% had no prior controls background.

Spring 2024 **Guest Lecturer**, *ME 450 Nonlinear Systems & Control*, Lehigh University, (Course Instructor: Prof. Eugenio Schuster).

- Delivered four lectures on Lyapunov theorem & feedback linearization.

Research Supervision

2022–Present **Graduate Student Supervisor**, *Lehigh University*, under Prof. Eugenio Schuster.
- Oversaw Ph.D. students' research on tokamak scenario optimization, plasma control, and AI-driven fusion solutions, fostering independent investigation and research skills.
- Managed project timelines to ensure milestones were met on schedule and student work aligned with research objectives.
- Advised on article preparation, conference presentations design to support research dissemination.

Pedagogical Training

Fall 2024 **Participant**, *Teacher Development Program, Level 1*, Lehigh University, Center for Innovation in Teaching and Learning.
- Completed comprehensive teacher development program focused on innovative teaching methodologies and pedagogical practices.

Teaching Support Roles

Fall 2019 – Spring 2020 **Graduate Teaching Assistant**, *Engineering Education*, Virginia Tech.
- Course: ENGE 1215, 1216: Foundations of Engineering.
- Number of students: ≈ 140.

Fall 2018 – Spring 2019 **Head Graduate Teaching Assistant**, *Mechanical Engineering*, Virginia Tech.
- Course: ME 4015, 4016: Engineering Design and Project.
- Number of students: ≈ 400. In charge of ≈ 70 students.

Fall 2017 – Spring 2018 **Graduate Teaching Assistant**, *Mechanical Engineering*, Virginia Tech.
- Course: ME 4015, 4016: Engineering Design and Project.
- Number of students: ≈ 400. In charge of ≈ 70 students.

Fall 2016 **Grader**, *System Dynamics, Mechanical Engineering*, Virginia Tech.

Spring 2016 **Grader**, *Differential Equations, Mathematics*, Virginia Tech.

Mentoring Experience

Summer 2017 & 2018 **Academic Coach**, *Student Transition Engineering Program*, Virginia Tech.

Summer, Fall 2018 **Peer Mentor**, *Virginia Tech's Early Engineering Mentoring program*, Virginia Tech.

Industry Work Experience

Sep 2014 - May 2015 **Design Engineer**, *Amritha Tool Crafts Pvt. Ltd.*, India, Full-time employee.

Academic Service

Working Groups, Panels, Workshops

2025 **Review Panelist**, *National Science Foundation (NSF)*, Directorate for Engineering (ENG), Division of Electrical, Communications and Cyber Systems (ENG/ECCS).

Jan 9-11, 2024 **Subject Matter Expert**, *Basic Research Needs Workshop on Measurement Innovation*, Fusion Pilot Plant (FPP)-Magnetic confinement Fusion Energy (MFE) working group.
Sponsored by the U.S. Department of Energy (DOE), Office of Science (SC), Office of Fusion Energy Sciences (FES).

Conference Sessions Organized and Chaired

2025 **Session Organizer**, *IEEE Conference on Control Technology and Applications*, San Diego, USA.
Invited Session: Optimization and Control Techniques in Nuclear Fusion and Plasma Science

2024 **Session Chair**, *IEEE American Control Conference 2024*, Toronto, Canada.
Regular Session: Nonlinear Systems Identification

Peer Review

2017 – ongoing **Peer Reviewer for Journals**.
Neurocomputing, Plasma Physics and Controlled Fusion, Neural Networks, Annual Reviews in Control, Nonlinear Dynamics, Fusion Engineering and Design, IEEE Control Systems Letters, International Journal of Robust and Nonlinear Control, International Journal of Adaptive Control and Signal Processing, Journal of the Franklin Institute, IEEE Control Systems Letters, Automatica, IEEE Transactions on Automatic Control, and MPDI Mathematics

2017 – ongoing **Peer Reviewer for Conferences**.
IEEE CDC, IEEE ACC, IEEE CCTA, IFAC SSSC, ASME SMASIS, and IFAC World Congress.

Professional Memberships

2017, 2020, 2022 – ongoing **Member**, *Institute of Electrical and Electronics Engineers (IEEE)*.

2021 – ongoing **Member**, *American Physical Society (APS)*.